Data Availability

The excitation dataset, the load-cell reference measurements and the complete analysis code are openly available at [**TODO:** repository and DOI], released under [**TODO:** licence].

The archive contains the 140 gate-passing excitation runs of Sec. VIII-A, covering five centre-of-mass configurations on two axes in both tip directions at four commanded ramp rates. Each run carries the three sensing channels used here — individual rotor speeds, from which the applied moment and collective thrust are reconstructed; the onboard rate gyro; and the motion-capture pose — together with the identified onset and the gate outcomes that produced it. The load-cell offset measurement that serves as ground truth is included for all ten case/axis configurations, as are the two-stage calibration constants and the commanded moment offsets flown in Sec. VIII-E.

Three uses are anticipated beyond reproducing the results reported here, and the archive is organised for them.

As a benchmark for onset detection. Every run is paired with an independent measurement of the quantity the onset is used to estimate, so a detector can be scored on the delivered offset and not only on self-consistency. The seven detectors compared in Table 2 are included with the harness that runs them, so a new method can be added and evaluated against the same ten configurations without re-deriving the inversion.

As a record of compliant ground contact. The motion-capture arcs resolve the pivot arm to a few millimetres and bound the rotation-centre height and the contact migration during the tip, quantities that are otherwise difficult to obtain for a landing gear under load. Sec. VI-B uses them to bound the ground-contact budget; they are reported in full so that a different contact model can be fitted to them.

As a testbed for tip-over dynamics. The ramp rate spans a factor of twelve, which separates the rate-dependent and excursion-dependent terms of any model of the departure, not only the one used here. The raw traces are provided so that alternative dynamic models can be identified from the same events.

One limitation should be stated with the release. The excitation was performed at a single rotor clearance, so the aerodynamic and contact-geometry contributions to the static threshold deficit are degenerate in this dataset and cannot be separated by any analysis of it; Sec. VI-B reports three independent attempts and the reason each fails. Separating them requires a clearance sweep, which the archive does not contain.

[**TODO:** state the archive size and format, e.g. "The archive is N GB; raw logs are provided as rosbags with a Parquet mirror of the derived per-run quantities."]